

Week 3

Data Fetching and Cleaning from Public Repositories

GEO, NCBI, and Reproducible R Workflows

Bioinformatics: From Multi-Omics Data to Discovery

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Two 40-Minute Sessions | Fall 2026

b Statistical modelling between features

Features can be related between modalities by modelling a statistical relationship

c Latent space inference

Mapping

Latent model prediction

Latent space

Mapping

Latent model prediction

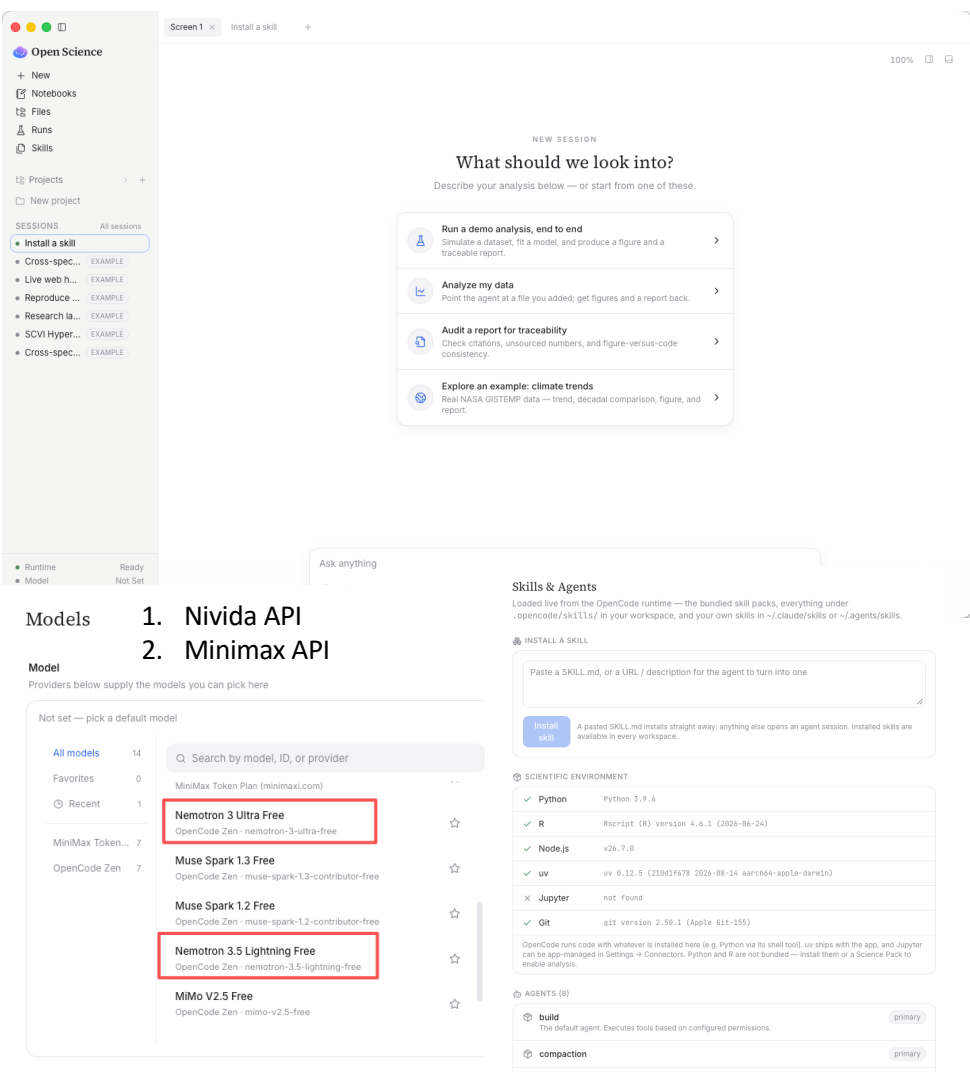
d Consensus of individual inferences (late integration)



安装包下载链接: <https://github.com/ai4s-research/open-science/releases/tag/v0.5.1>

Open.Science-0.5.1-1.x86_64.rpm		sha256:2bd5a119bd1c96aaef7d7caca0d...		113 MB	last week
Open.Science_0.5.1_aarch64.dmg	MacOS	sha256:bb1ab93f07a73a4c6f928c96a4...		106 MB	last week
Open.Science_0.5.1_amd64.deb		sha256:4656b8def814323e4350b2ed27...		113 MB	last week
Open.Science_0.5.1_x64-setup.exe	Windows	sha256:03c83429d8e217f4d70684eecb...		77.3 MB	last week
Open.Science_0.5.1_x64.dmg		sha256:ac10d8225ade70e943e84e1d37...		114 MB	last week
Open.Science_0.5.1_x64_en-US.msi		sha256:e21727fe6174564807e248ff9e...		108 MB	last week
Open.Science_aarch64.app.tar.gz		sha256:c8550f6abecb7450cd9a9161fa...		90.3 MB	last week
Open.Science_x64.app.tar.gz		sha256:4812fe3795112d9e23b932aa10...		97.1 MB	last week
osd-0.5.1-aarch64-apple-darwin.tar.gz		sha256:84e3aab3e48c189406b40bf918...		80.6 MB	last week
osd-0.5.1-x86_64-apple-darwin.tar.gz		sha256:d32dab117b6c9c5383e6a5ea8a...		84.5 MB	last week
osd-0.5.1-x86_64-pc-windows-msvc.zip		sha256:4dea2eeca766706e9ac32c35cc...		95.8 MB	last week
osd-0.5.1-x86_64-unknown-linux-gnu.tar.gz		sha256:dd15165111f459a4b4cc3440eb...		98.4 MB	last week
Source code (zip)					last week
Source code (tar.gz)					last week

👍 5 people reacted



Learn
Analyze
Create
Grow

Different analysis pathways for different student skill levels

From no-code tools to AI-agent-assisted projects

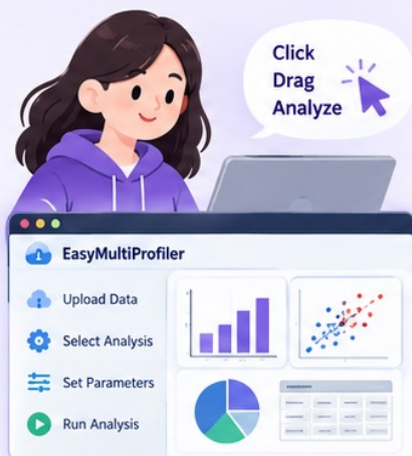
Same
Data
More Possibilities

1

Level 1

EasyMultiProfiler-web

No-code



Basic homework

2

Level 2

EasyMultiProfiler in R

Template-based



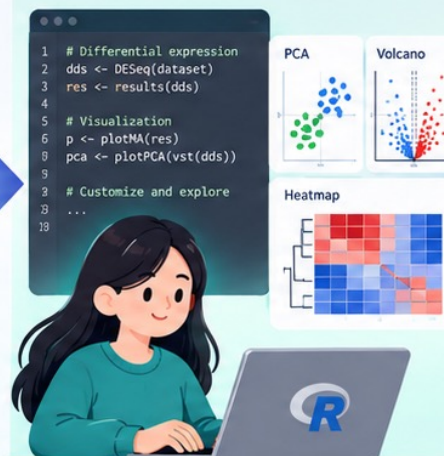
Guided assignment

3

Level 3

R programming

Flexible analysis



Independent project

4

Level 4

R + AI agents

Agent-assisted



Advanced project



Ease of use



Customization



Reproducibility



Automation

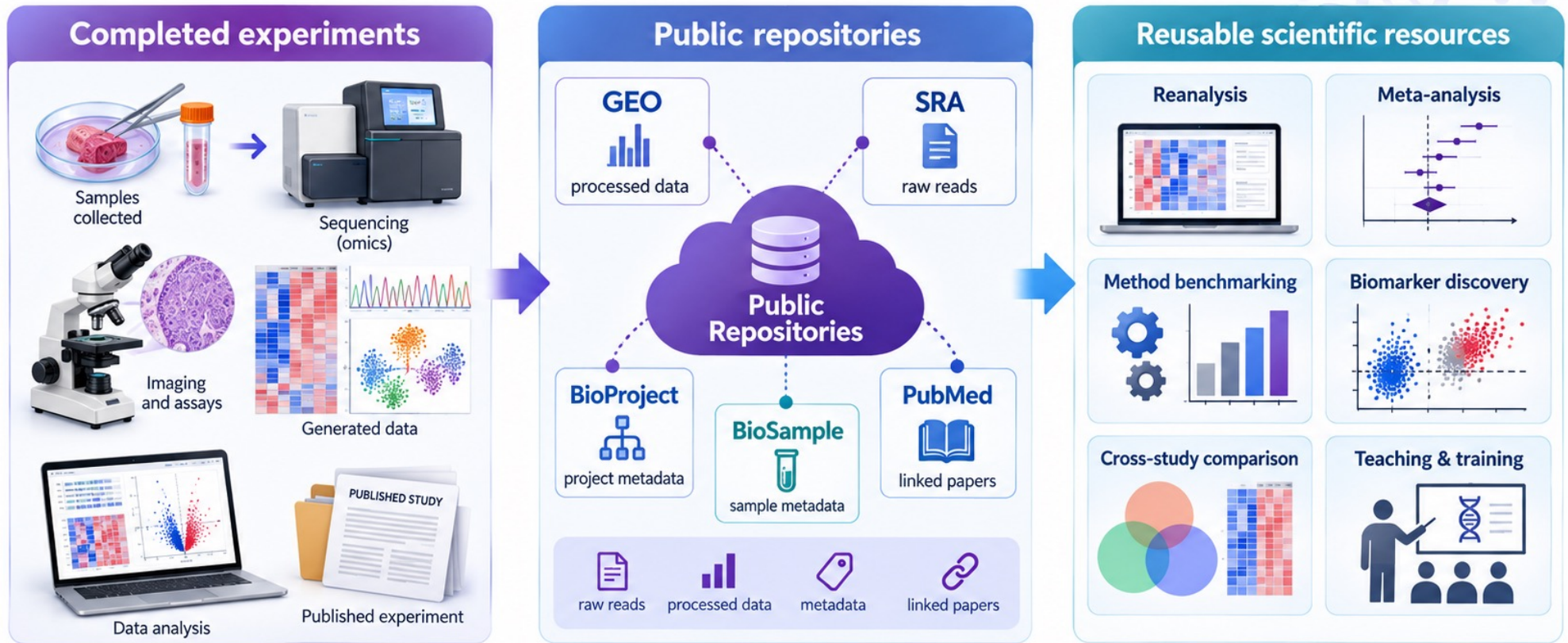
Higher
Impact



Data Skills Today
A Healthier Tomorrow

BIOINFORMATICS • STUDENTS • BIGGER TOMORROWS

Public repositories turn completed experiments into reusable scientific resources

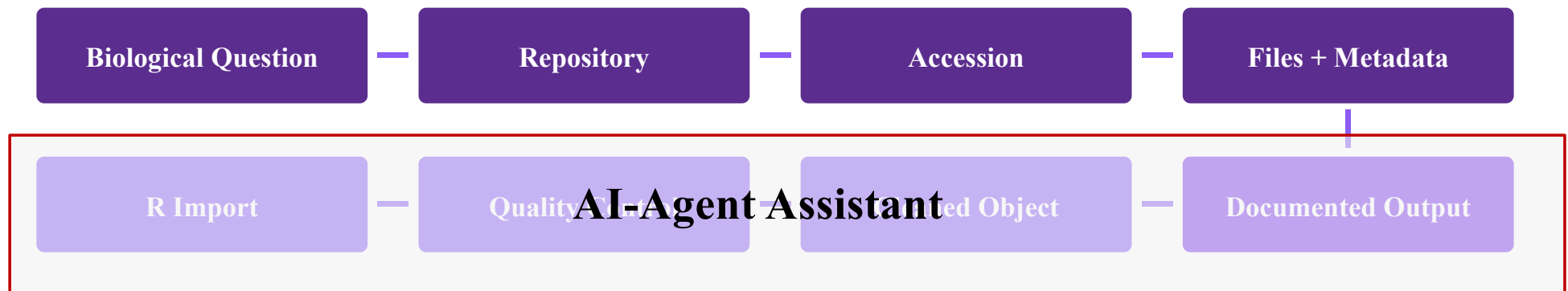


Sharing data and metadata enables **validation**, **discovery**, and **new biological insight** beyond the original study.



Week 3 moves from repository navigation to reproducible data cleaning

Learning Architecture



By the End of Week 3, You Will Be Able To

Navigate GEO and NCBI repositories | Evaluate dataset suitability for a biological question

Retrieve data reproducibly with GEOquery | Inspect, clean, and document an analysis-ready object

Two classes combine repository literacy with practical R workflows

Class 1

Public Repositories and Dataset Selection

Focus: Navigate NCBI / GEO, read a Series page, judge dataset quality

Activity: Accession matching + dataset triage

Output: Exit ticket on repository literacy

Class 2

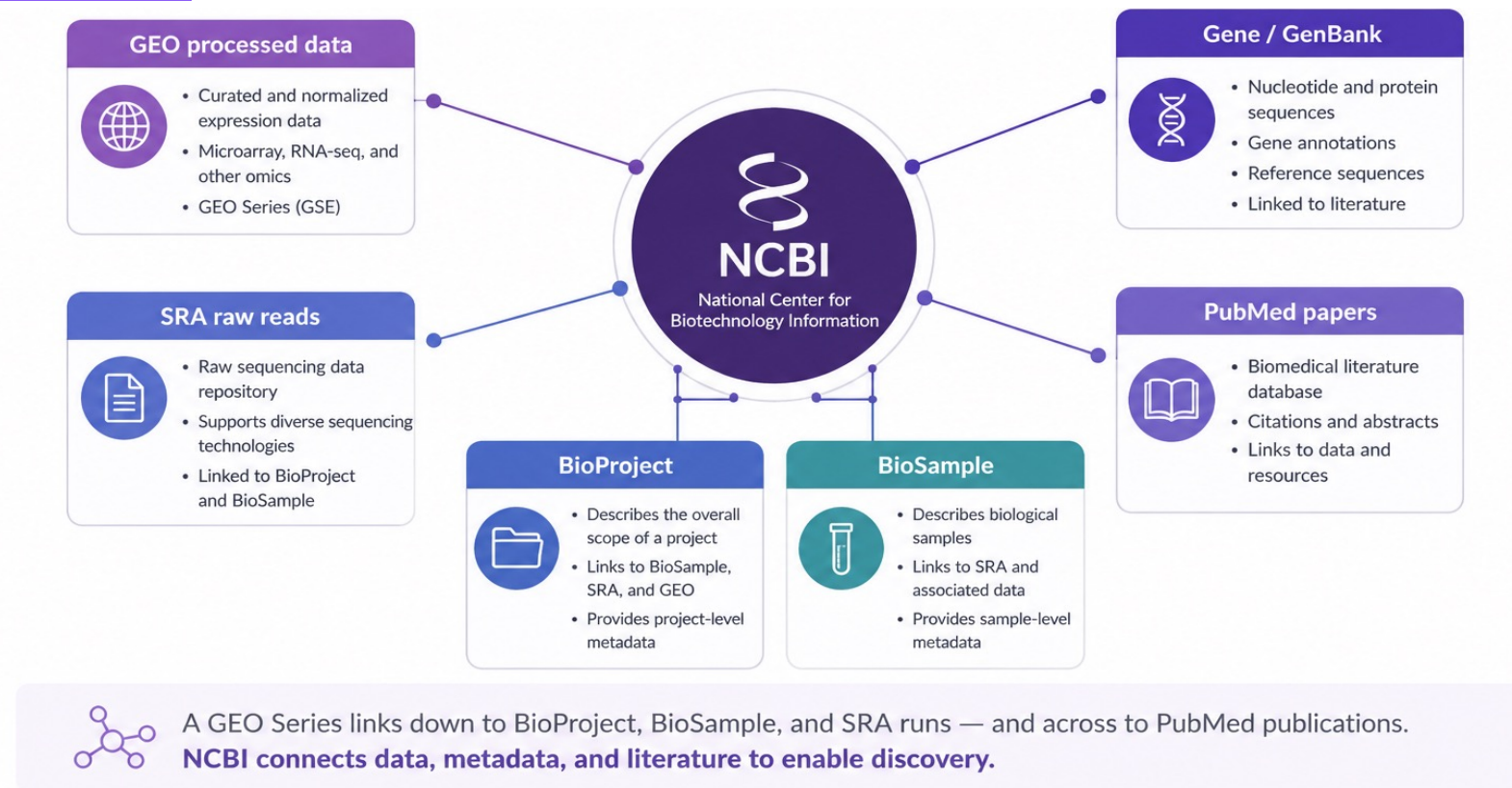
Retrieval, Cleaning, QC, and AI Assistance

Focus: GEOquery, inspection, cleaning, PCA, batch, normalization

Activity: AI code error-detection clinic

Output: One-hour homework + quiz prep

NCBI is an interconnected ecosystem rather than a single database



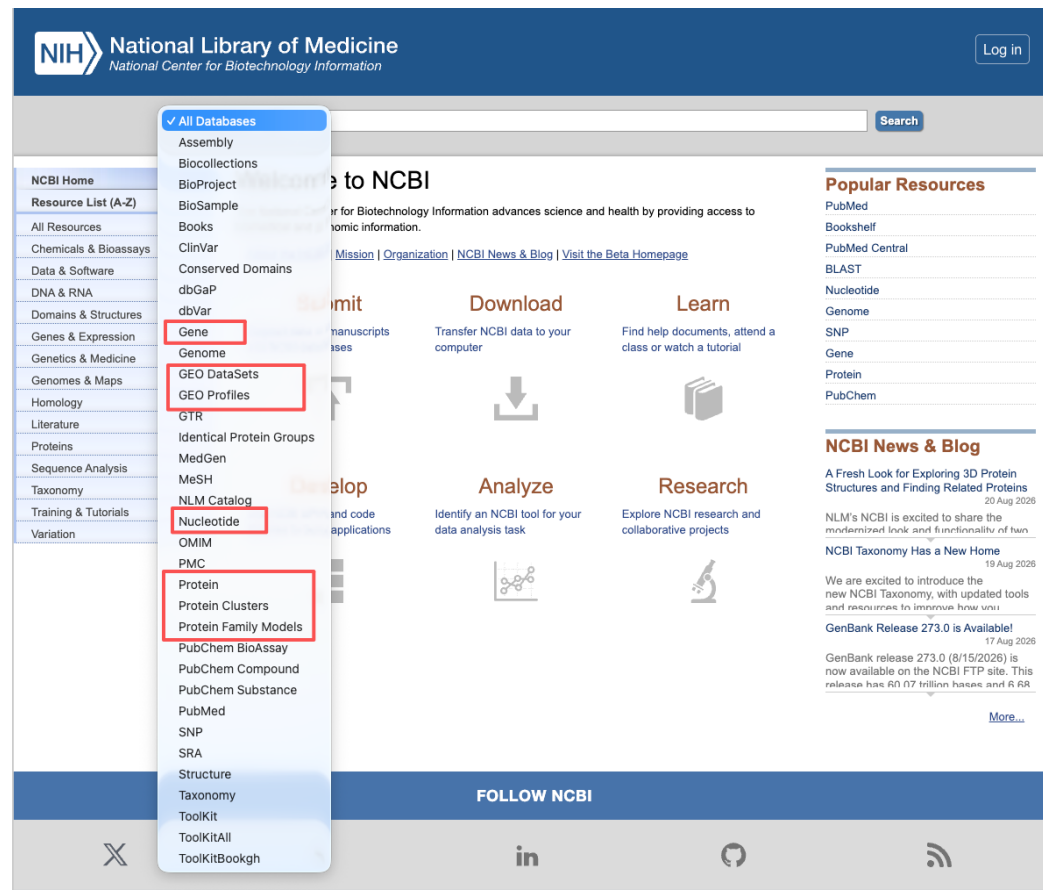
NCBI is an interconnected ecosystem rather than a single database

<https://www.ncbi.nlm.nih.gov/>

A section from the reading material for homework from the last week

Data availability

Raw data for host RNA-seq for the CRC cohort are available on the NCBI Sequence Read Archive (SRA) under BioProject ID [PRJNA816986](#). Raw data for previously published 16S rRNA sequencing for the CRC cohort can be accessed at [PRJNA284355](#)⁵. Raw data for previously published 16S rRNA sequencing and host RNA-seq for the IBD cohort can be accessed at [PRJNA398089](#) and [GSE111889](#), respectively^{22,25}. Raw data for 16S rRNA sequencing and host RNA-seq for the IBS cohort can be accessed at [PRJEB37924](#) and [GSE146853](#), respectively⁸. Processed data tables for host transcriptomics and microbiome data for each disease cohort have been included as supplementary tables (Supplementary Tables 12–17).



NCBI is an interconnected ecosystem rather than a single database

All Databases PRJNA816986 Search

BIOPROJECT
Shared and disease-specific host gene-microbiome interactions across human diseases
Raw host RNA-seq data from biopsy samples collected from patients with colorectal cancer. The goal of this study was to study interactions between host gene expression and gut microbiome across human gastrointestinal diseases, including colorectal ...
PRJNA816986

Literature

Bookshelf 0
MeSH 0
NLM Catalog 0
PubMed 0
PubMed Central 3

Genomes

Assembly / Genome NCBI Datasets
BioCollections 0
BioProject 1
BioSample 0
Nucleotide 0
SRA 88
Taxonomy 0

Genes

Gene 0
GEO DataSets 0
GEO Profiles 0

Clinical

ClinicalTrials.gov 0
ClinVar 0
dbGaP error
dbSNP 0
dbVar 0
GTR 0
MedGen 0
OMIM 0

Project Data:

Resource Name	Number of Links
SRA Experiments	88
BioSample	88

SRA Data Details

Parameter	Value
Data volume, Gbases	205
Data volume, Mbytes	92764

Access

Public (88)
Source RNA (88)
Library Layout paired (88)
Platform Illumina (88)
Strategy other (88)
Data in Cloud GSI (88)
File Type fastq (88)
Clear all
Show additional filters

Summary 20 per page
View results as an expanded interactive table using the RunSelector. Send results to Run selector
Links from BioProject
Items: 1 to 20 of 88

- RNA-seq from colorectal cancer patients
- RNA-seq from colorectal cancer patients
- RNA-seq from colorectal cancer patients
- RNA-seq from colorectal cancer patients
- RNA-seq from colorectal cancer patients

Accession: PRJNA816986 ID: 816986

Shared and disease-specific host gene-microbiome interactions across human diseases
Raw host RNA-seq data from biopsy samples collected from patients with colorectal cancer. More...

Accession: PRJNA816986

Data Type Raw sequence reads

Scope Multispecies

Grants "Population Genomics of Host-Microbiome Interactions" (Grant ID R35 GM128716, National Institute of General Medical Sciences)

Submission Registration date: 16-Mar-2022
University of Minnesota

Relevance Microbiome and host transcriptomics in human diseases

BioSample

Summary 20 per page Sort by Default order Send to: *

Organism Customize ...

Attribute name disease isolate sex tissue Customize ...

Sample type

Access Public (88)
Other Used by SRA (88)
Clear all
Show additional filters

Links from BioProject
Items: 1 to 20 of 88

- Human sample from Homo sapiens
- Human sample from Homo sapiens
- Human sample from Homo sapiens
- Human sample from Homo sapiens
- Human sample from Homo sapiens

NCBI is an interconnected ecosystem rather than a single database

NCBI

GEO
Gene Expression Omnibus

HOME | SEARCH | SITE MAP

GEO Publications | FAQ | MIAME | Email GEO

NCBI > GEO > Accession Display

Not logged in | Login

https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi

Scope: Self | Format: HTML | Amount: Quick | GEO accession: GSE111889

Series GSE111889

Query DataSets for GSE111889

Status

Public on Mar 17, 2018

Title

Longitudinal Multi-omics of the Human Microbiome in Inflammatory Bowel Disease

Organism

Homo sapiens

Experiment type

Expression profiling by high throughput sequencing

Summary

Purpose: The main Inflammatory Bowel Disease (IBD) Multi-omics Database (IBDMDB) study includes multi-omics measurements from over 100 subjects, sampled biweekly over up to a year in both adult and pediatric patients with IBD (Crohn's disease and ulcerative colitis), along with non-IBD controls. Data types include fecal metagenomes, metatranscriptomes, metabolomes, and proteomes, as well as host genetics, intestinal biopsy transcriptomes, epigenetics, and 16S amplicon profiles. Subjects' medical histories and demographics are collected at baseline and medication, diet, and disease activity profiled longitudinally.

Methods: Methods: Total RNA was quantified using the Quant-IT[™] RiboGreen[®] RNA Assay Kit and normalized to 5ng/ul. Following plating, 2 uL of ERCC controls (using a 1:1000 dilution) were spiked into each sample. An aliquot of 200ng for each sample was transferred into library preparation which was an automated variant of the Illumina TruSeq[™] Stranded mRNA Sample Preparation Kit. This method preserves strand orientation of the RNA transcript. It uses oligo dT beads to select mRNA from the total RNA sample. It is followed by heat fragmentation and cDNA synthesis from the RNA template. The resultant 500bp cDNA then goes through library preparation (end repair, base 'A' addition, adapter ligation, and enrichment) using Broad designed indexed adapters substituted in for multiplexing. After enrichment the libraries were quantified using Quant-IT PicoGreen (1:200 dilution). After normalizing samples to 5 ng/uL, the set was pooled and quantified using the KAPA Library Quantification Kit for Illumina Sequencing Platforms. The entire process is in 96-well format and all pipetting is done by either Agilent Bravo or Hamilton Starlet. Pooled libraries were normalized to 2nM and denatured using 0.1 N NaOH prior to sequencing. Flowcell cluster amplification and sequencing were performed according to the manufacturer's protocols using either the HiSeq 2000.

Overall design

RNA-seq analysis of biopsy samples retrieved from CD, UC and non-IBD patients enrolled in the HMP2 project.
NOTE: Submitter declares that the raw data will be submitted to dbGaP (https://www.ncbi.nlm.nih.gov/gap) because of patient privacy concerns.

Contributor(s)

Arze C, Plichata D, Bishai J

Citation(s)

Lloyd-Price J, Arze C, Ananthakrishnan AN, Schirmer M et al. Multi-omics of the gut microbial ecosystem in inflammatory bowel diseases. *Nature* 2019 May;569(7758):655-662. PMID: 31142855^{PMID} IF: 56.1^{Q1}

Submission date

Mar 15, 2018

Last update date

Jun 04, 2019

Contact name

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Department

Biostatistics

Lab

Huttenhower

Street address

677 Huntington Ave

City

Boston

State/province

MA

ZIP/Postal code

02115

Country

USA

Platforms (1)

GPL11154 Illumina HiSeq 2000 (Homo sapiens)

Samples (251)

More...

GSM3043377 CSM5FZ1G

GSM3043378 CSM5FZ1F

GSM3043379 CSM5FZ1I

Relations

BioProject

PRJNA438663

Download family

SOFT formatted family file(s)

MINI ML formatted family file(s)

Series Matrix File(s)

Format

SOFT

MINI ML

TXT

Supplementary file

Size

Download

File type/resource

GSE111889_RAW.tar

68.5 Mb

(http)(custom)

TAR (of TSV)

GSE111889_host_tx_counts.tsv.gz

10.6 Mb

(ftp)(http)

TSV

Raw data are available in SRA

Processed data provided as supplementary file

Processed data are available on Series record

Supplementary file

Size

Download

File type/resource

GSE111889_RAW.tar

68.5 Mb

(http)(custom)

TAR (of TSV)

GSE111889_host_tx_counts.tsv.gz

10.6 Mb

(ftp)(http)

TSV

Raw data are available in SRA

Processed data provided as supplementary file

Processed data are available on Series record

Custom GSE111889_RAW.tar archive:

Supplementary file

File size

☐ GSM3043377_CSM5FZ1G.tsv.gz

274.8 Kb

☐ GSM3043378_CSM5FZ1F.tsv.gz

203.3 Kb

☐ GSM3043379_CSM5FZ1I.tsv.gz

277.0 Kb

☐ GSM3043380_CSM5FZ1H.tsv.gz

276.5 Kb

☐ GSM3043381_CSM5FZ1J.tsv.gz

279.3 Kb

☐ GSM3043382_CSM5FZ1K.tsv.gz

279.1 Kb

☐ GSM3043383_CSM5FZ1L.tsv.gz

277.2 Kb

☐ GSM3043384_CSM5FZ1O.tsv.gz

276.0 Kb

☐ GSM3043385_CSM5FZ1M.tsv.gz

277.2 Kb

☐ GSM3043386_CSM5FZ1N.tsv.gz

277.9 Kb

☐ GSM3043387_CSM5FZ1Q.tsv.gz

281.0 Kb

☐ GSM3043388_CSM5FZ1R.tsv.gz

280.7 Kb

☐ GSM3043389_CSM5FZ1S.tsv.gz

277.5 Kb

☐ GSM3043390_CSM5FZ1W.tsv.gz

281.3 Kb

☐ GSM3043391_CSM5FZ1T.tsv.gz

276.3 Kb

☐ GSM3043392_CSM5FZ1U.tsv.gz

283.0 Kb

☐ GSM3043393_CSM5FZ1V.tsv.gz

280.0 Kb

☐ GSM3043394_CSM5FZ22.tsv.gz

279.3 Kb

☐ GSM3043395_CSM5FZ24.tsv.gz

279.4 Kb

☐ GSM3043396_CSM5FZ23.tsv.gz

279.0 Kb

☐ Select All

Cancel

Download

0 file(s), 0 b

The correct repository depends on data type and analysis goal

Repository	Primary Content	Typical Use	Access / Format
GEO	Processed expression + sample metadata	Download analyzed matrices	Series Matrix, TXT, SOFT
SRA / ENA	Raw sequencing reads (FASTQ)	Re-alignment, variant calling	FASTQ, SRA toolkit
TCGA / GDC	Cancer multi-omics + clinical	Tumor atlas analysis	Portal, API, controlled
GTEx	Normal human tissue expression	Baseline tissue comparison	Portal, open + controlled
ENCODE	Functional genomics, ChIP/ATAC	Regulatory element mapping	Portal, BAM/BED/bigWig
PRIDE	Proteomics mass spectrometry	Protein identification / quant	RAW, mzML, mzTab
Mgnify / NMDC	Microbiome sequences + analysis	Metagenomic profiling	Portal, API, FASTA

GEO and SRA often represent different layers of the same experiment

GEO Series (GSE)

The processed layer — ready for analysis

- Processed expression matrices
- Sample metadata tables
- Platform (GPL) annotation

SRA (SRR runs)

The raw layer — requires processing

- Raw FASTQ sequencing reads
- Linked via BioProject / BioSample
- Needs alignment + quantification



Key Distinction

GEO gives you someone else's processed result. SRA gives you the raw material. **Your analysis plan decides which layer you need.**

GEO accession prefixes encode the study hierarchy

GSE — Series (the whole study)

e.g., GSE158422 — summary, design, files, links

GSM — Samples (one per row)

e.g., GSM4799021 — per-sample metadata + data links

GPL — Platform (the technology)

e.g., GPL24676 — probe / feature annotation table

Supplementary Files

Where the actual data live: count matrices, normalized tables, or links to raw reads

Also seen: GDS = curated GEO DataSet | SRR = SRA run | PRJNA = BioProject | SAMN = BioSample

Bio-database for USA and Europe

<https://www.ncbi.nlm.nih.gov/>

The screenshot shows the NCBI homepage with a dark blue header containing the NIH logo and 'National Library of Medicine' text. Below the header is a search bar with a dropdown menu set to 'All Databases' and a 'Search' button. The main content area is divided into three columns. The left column is a 'Resource List (A-Z)' with links to various databases. The middle column is titled 'Welcome to NCBI' and features six large icons with labels: 'Submit' (Deposit data or manuscripts into NCBI databases), 'Download' (Transfer NCBI data to your computer), 'Learn' (Find help documents, attend a class or watch a tutorial), 'Develop' (Use NCBI APIs and code libraries to build applications), 'Analyze' (Identify an NCBI tool for your data analysis task), and 'Research' (Explore NCBI research and collaborative projects). The right column is titled 'Popular Resources' and lists links to PubMed, Bookshelf, PubMed Central, BLAST, Nucleotide, Genome, SNP, Gene, Protein, and PubChem. Below this is the 'NCBI News & Blog' section, which includes a 'Fresh Look for Exploring 3D Protein Structures and Finding Related Proteins' article dated 20 Aug 2026, a 'NCBI Taxonomy Has a New Home' article dated 19 Aug 2026, and a 'GenBank Release 273.0 is Available!' article dated 17 Aug 2026. A 'More...' link is at the bottom of the news section.

<https://www.ebi.ac.uk/>

The screenshot shows the EBI Search results page. The header includes the EMBL logo and 'EMBL's European Bioinformatics Institute' text. The main search bar contains the query 'PRJNA816986' and a 'Search' button. Below the search bar, it says 'Search results for PRJNA816986' and 'Showing 15 results out of 178 in All results'. A link to 'Give us feedback on these results' is provided. The results are filtered by 'Nucleotide sequences (178 results)'. The first result is 'PRJNA816986' from 'Source: ENA Study (ID: PRJNA816986)', described as 'Shared and disease-specific host gene-microbiome interactions across human diseases'. It has 'Cross references: Nucleotide sequences (177)'. The second result is 'SRP364890' from 'Source: Study (Read/Analysis) (ID: SRP364890)', described as 'Shared and disease-specific host gene-microbiome interactions across human diseases'. It has 'Cross references: Nucleotide sequences (177)'. The third result is 'SRR18371922' from 'Source: Read (Run) (ID: SRR18371922)', described as 'Illumina HiSeq 2500 sequencing: RNA-seq from colorectal cancer patients'. On the right side of the page, there is a 'Train with us' button and a banner for the 'Antimicrobial resistance portal' with the text 'AMR Portal Release 2026_07 is live' and the date '18 Aug 2026'.

A GEO Series page contains the clues needed to reconstruct the experiment

GEO Series GSE158422 (mockup)

Title	RNA-seq of colorectal tumors and matched normals
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Summary

Study rationale and main conclusion

Design Overall design: 40 tumor vs 40 matched normal

Organism Homo sapiens

Platform GPL24676 Illumina NovaSeq 6000

Samples GSM4799021 ... GSM4799100 (80 samples)

Files Supplementary: raw_counts.txt.gz, metadata.xlsx

Citation PubMed ID 33442210 — linked publication

Series	GSE111889	Query	Datasets for GSE111889
Status	Public on Mar 17, 2018		
Title	Longitudinal Multioleomics of the Human Microbiome in Inflammatory Bowel Disease		
Organism	<i>Homo sapiens</i>		
Experiment type	Expression profiling by high throughput sequencing		
Summary	Purpose: The main Inflammatory Bowel Disease (IBD) Multioleomics Database (IBDMDDB) study includes multioleomics measurements from over 100 subjects, sampled biweekly over up to a year in both adult and pediatric patients with IBD (Crohn's disease and ulcerative colitis), along with non-IBD controls. Data types include fecal metagenomes, metatranscriptomes, metabolomes, and proteomes, as well as host genetics, intestinal biopsy transcriptomes, epigenetics, and 16S amplicon profiles. Subjects' medical histories and demographics are collected at baseline and medication, diet, and disease activity profiled longitudinally. Methods: Methods: Total RNA was quantified using theQuant-iT[®] RiboGreen[®] RNA Assay Kit and normalized to 5ng/ul. Following plating, 2 ul of ERCC controls (using a 1:1000 dilution) were spiked into each sample. An aliquot of 200ng for each sample was transferred into library preparation which was an automated variant of the Illumina TruSeq[®] Stranded mRNA Sample Preparation Kit. This method preserves strand orientation of the RNA transcript. It uses oligo dT beads to select mRNA from the total RNA sample. It is followed by heat fragmentation and cDNA synthesis from the RNA template. The resultant 500bp cDNA then goes through library preparation (end repair, base 'A' addition, adapter ligation, and enrichment) using Broad designed indexed adapters substituted in for multiplexing. After enrichment the libraries were quantified using Quant-iT PicoGreen (1:200 dilution). After normalizing samples to 5 ng/ul, the set was pooled and ligated using the KAPA Library Quantification Kit for Illumina Sequencing Platforms. The entire process is in 96-well format and all pipetting is done by either Agilent Bravo or Hamilton Starlet. Pooled libraries were normalized to 2nM and denatured using 0.1 N NaOH prior to sequencing. Flowcell cluster amplification and sequencing were performed according to the manufacturer's protocols using either the HiSeq 2000.		
Overall design	RNA-seq analysis of biopsy samples retrieved from CD, UC and non-IBD patients enrolled in the HMP2 project. NOTE: Submitter declares that the raw data will be submitted to dbGaP (https://www.ncbi.nlm.nih.gov/gap) because of patient privacy concerns.		
Contributor(s)	Arze C, Plichta D, Bishal J		
Citation(s)	Lloyd-Pierce J, Arze C, Ananthakrishnan AN, Schirmer M et al. Multi-omics of the gut microbial ecosystem in inflammatory bowel diseases. <i>Nature</i> 2019 May;369(7758):655-662. PMID: 31142853		
Submission date	Mar 15, 2018		
Last update date	Jun 04, 2019		
Contact name	Curtis Huttenhower		
Organization name	Harvard T.H. Chan School of Public Health		
Department	Biostatistics		
Lab	Huttenhower		
Street address	677 Huntington Ave		
City	Boston		
State/province	MA		
ZIP/Postal code	02115		
Country	USA		

EN

European Nucleotide Archive

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Project: PRJNA498663

Purpose: The PRJNA498663 database (BIC Mutomic Database (BICMDB)) stores multiric measurements from over 130 subjects who were followed up at a year interval after initial positive patients with BIC diagnosis and operative counts, along with non-BIC controls. Data types include fold misregulations, metabolite/proteins, and proteomics, as well as host genes' differential gene expression, transcriptomics, and 16S amplicon profile. Subject's medical histories and demographic are collected at baseline and follow-up, and disease activity profile longitudinally. Methods: Method: Total RNA was quantified using RiboZero[®] (RZ) Ribosomal RNA Akit and normalized to quantity. Following pipette 1 µL of cDNA controls (using a 1:100 dilution) were added into each sample. An aliquot of 200µg per each sample was transferred into library preparation which was an automated version of the Illumina TruSeq[®] Stranded mRNA Sample Preparation kit. This method provides strand orientation of the RNA transcripts. A size range of 150bp to select mRNA from the total RNA prep. It is followed by bead fragmentation and cDNA synthesis from the first template. The resultant cDNA-free RNA free space through library preparation and ligated. As a addition, adapter ligation, and enrichment using bead designed oriented adapters is performed for multiplexing. After enrichment the libraries were quantified using Qubit[®] Fluorometer (1:200 dilution). After normalizing samples to 10 ng/µL, the set was pooled and sequenced using the KAPA Library Quantification kit for Illumina Sequencing Platforms. The entire process is in 96-well format and all pipetting is done by either Agilent Bravo or Hamilton Starline. Pooled libraries were submitted to 2nd end sequencing using 150 bp paired-end sequencing. Paired-end analysis and sequencing were performed according to the manufacturer's instructions. After the HISEQ 2500, overall reads were analyzed of mapping rates between reads, GC bias, and non-BIC pairs.

Note: The project has been approved by the Institutional Review Boards (IRBs) of the University of California, San Diego and the National Cancer Institute. All data will be submitted to public databases (<http://www.ncbi.nlm.nih.gov/bioproject/>) once the data have been fully processed.

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Addicted to clipboard

General

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- > Publications: [Show](#)
- > Parent Projects: [Show](#)
- > ORCID Data Records: [Show](#)
- > Related ENA Records: [Hide](#)

Data

This public data is associated with this project

Organism: [Homo sapiens \(human\)](#)

Study Title: [Longitudinal Multi-omics of the Human Microbiome in Inflammatory Bowel Disease](#)

Contact Name: [Hughesmore, Biostatistics, Harvard T.H. Chan School of Public Health](#)

Study Name: [Longitudinal Multi-omics of the Human Microbiome in Inflammatory Bowel Disease](#)

ENA-REFSEQ: [X](#)

PROJECT-ID: [X](#)

ENA-FIRST-PUBLIC: [2014-03-18](#)

ENA-LAST-UPDATE: [2020-09-03](#)

Related ENA Records

Project	Count
PRJNA498663	1

Metadata determine whether a matrix can answer a biological question

Expression Matrix

Gene	GSM001	GSM002	GSM003	GSM004
LDHA	8452	1203	9104	1187
MKI67	5210	980	6043	1042
IL6	312	95	428	88
ACTB	15230	14890	16102	15045



Phenotype Metadata

Sample	Group	Age	Batch
GSM001	Tumor	62	B1
GSM002	Normal	58	B1
GSM003	Tumor	67	B2
GSM004	Normal	61	B2

Why Metadata Matter

The matrix alone cannot answer "tumor vs normal". Only the metadata link each column to its biology — **group, tissue, age, sex, batch** — enabling both the comparison and the confounder control.

Sample identifiers must match exactly across expression and metadata tables

Correct Alignment

Matrix column	Metadata row	Match
GSM001	GSM001	Yes
GSM002	GSM002	Yes
GSM003	GSM003	Yes

Silent Misalignment

Matrix column	Metadata row	Match
GSM001	GSM002	Shifted
GSM002	GSM003	Shifted
GSM003	GSM001	Shifted



The Alignment Rule

Never assume two files share the same sample order. Always verify with **identical(colnames(expr), rownames(meta))**, and align explicitly — a shifted table silently assigns tumor labels to normal samples.

Raw counts, processed values, and FASTQ files support different analyses

Data Level	Typical File	Supports	Caution
Raw integer counts	counts.txt.gz	DESeq2 / edgeR differential expression	Required by count-based models
Log microarray	series_matrix.txt	limma linear models, PCA, heatmaps	Check whether already log2-transformed
TPM / FPKM	tpm_matrix.csv	Visualization, cross-sample display	Not valid input for DESeq2
FASTQ reads	SRR runs via SRA	Re-alignment, variant calling, novel features	Requires full processing pipeline

Rule of thumb: The statistical model chooses the data level — not the other way around. Decide your analysis method first, then download accordingly.

Dataset selection is a quality-control decision made before downloading

Criterion	What to Check on the GEO Page
Question fit	Does the design actually address your question?
Organism & tissue	Correct species, tissue, and condition?
Sample size & balance	Replicates per group? Groups comparable?
Metadata completeness	Group, covariates, and batch annotated?
Data availability	Raw counts or valid processed matrix present?
Platform & batch info	Single platform? Run dates / lanes reported?

Red flags can make a dataset unsuitable despite an attractive title

Too Few Replicates

Fewer than 3 per group — no power for differential analysis

Unclear Groups

Condition labels missing or ambiguous in metadata

Mixed Tissues

Different tissues or platforms pooled without annotation

No Raw Counts

Only TPM available, but your plan needs DESeq2

Batch = Condition

All cases run in batch 1, all controls in batch 2

Inconsistent IDs

Matrix columns and metadata rows cannot be matched

When You See a Red Flag

First look for another dataset. If none exists, narrow your question to what the data can honestly support — and document the limitation.

Public data reuse requires a traceable provenance record

Provenance Card (keep with every project)

Accession: GSE158422

File: GSE158422_raw_counts.txt.gz

Downloaded: 2026-07-21 from GEO FTP

Checksum: md5 = a3f2... (verify integrity)

Publication: PMID 33442210

Local path: data_raw/GSE158422/

Cleaning decisions: logged in logs/cleaning.md

Exit Ticket: Class 1

1. Which repository and accession type do you need for your question?
2. Does the dataset provide the data level your method requires?
3. One red flag you would check first?

Class 1 takeaway: If you cannot reconstruct the experiment from the GEO page, you are not ready to analyze the data. Next: retrieve it reproducibly in R.

02

Reproducible Retrieval, Cleaning, and AI-Assisted Workflows

Class 2 | From Accession to Analysis-Ready Data

GEOquery in R | Inspection before analysis | Microarray vs RNA-seq cleaning

QC plots, PCA, batch effects | Normalization | AI-assisted verified coding

A reproducible project separates raw data, processed data, and documentation

week3_project/

```
|-- data_raw/          # downloaded, untouched
|   |-- GSE158422_counts.txt.gz
|-- data_processed/    # cleaned outputs (RDS)
|-- metadata/          # sample tables, scorecard
|-- scripts/           # 01_fetch.R 02_clean.R 03_qc.R
|-- figures/           # exported plots
|-- logs/              # decisions, prompt log, errors
|-- manifest.tsv       # file provenance record
|-- session_info.txt   # R + package versions
```

Three Working Rules

1. Never overwrite raw data. Every transformation writes a new file in data_processed/.

2. Scripts run top to bottom. Numbered scripts let anyone reproduce the pipeline in order.

3. Log every decision. Filtering thresholds, exclusions, and AI prompts go in logs/.

GEOquery provides a reproducible interface between GEO and R

01_fetch.R

```
library(GEOquery)

gse <- getGEO("GSE158422", GSEMatrix = TRUE)
eset <- gse[[1]]          # ExpressionSet

counts <- exprs(eset)     # genes x samples
meta   <- pData(eset)     # sample metadata
feature <- fData(eset)    # feature annotation

getGEOSuppFiles("GSE158422") # raw counts
saveRDS(eset, "data_processed/eset_raw.rds")
```

The ExpressionSet Object

exprs() — the numeric matrix: rows are genes, columns are samples.

pData() — phenotype table; its row order must match the matrix columns.

fData() — feature annotation mapping probe or gene IDs to symbols.

Save the object once — never re-download.

Before cleaning, inspect dimensions, identifiers, and metadata structure

First Look in R

```
dim(counts)           # 20000 genes x 12 samples?
colnames(counts)       # GSM accessions?
rownames(counts)[1:5]

                        # probe IDs or gene symbols?
head(counts[, 1:3])    # integers or decimals?
colnames(meta)         # which column holds group?
table(meta$group)      # balanced design?
all(colnames(counts) == rownames(meta))

                        # must be TRUE
```

What You Are Verifying

Scale. Does the gene and sample count match the GEO record?

Identifier type. Probes need mapping; symbols may be usable directly.

Data scale. Integers suggest raw counts; decimals suggest TPM or log values.

Alignment. Matrix columns and metadata rows must correspond exactly.

The cleaning path depends on which data level you downloaded

Raw Counts

Best case.

- Integer matrix, genes x samples
- Filter low counts
- Feed directly to DESeq2
- Full control of normalization

Normalized Matrix

Usable with care.

- Identify method from GEO page
- TPM / FPKM / log values
- Fine for PCA and heatmaps
- Not valid for DESeq2 input

Supplementary Only

Hardest case.

- Mixed file formats
- Parse and merge per sample
- Verify IDs are consistent
- Document every manual step

Decide the branch **before** writing code — record it in your log with the evidence.

Integrity checks catch silent errors before they corrupt your analysis

Check	R One-Liner	Failure Means
Duplicate genes	<code>anyDuplicated(rownames(x))</code>	ID mapping collision
Missing values	<code>sum(is.na(x))</code>	Parsing or join error
Dimension match	<code>ncol(x) == nrow(meta)</code>	Dropped samples
Order match	<code>all(colnames(x) == rownames(meta))</code>	Mislabeled groups
Value range	<code>range(x, na.rm = TRUE)</code>	Wrong data scale

Run all five checks on every new file — they cost seconds.

Why "Silent"?

R does not warn you when a merge reorders samples or a join drops genes.

The code runs. The plot renders. The answer is wrong.

Rule: `stopifnot()` every assumption. Let the script fail loudly, not silently.

Filtering removes uninformative genes and sharpens the signal

A Standard Low-Count Filter

```
# keep genes with >= 10 counts in >= 3 samples
keep <- rowSums(counts >= 10) >= 3
counts_f <- counts[keep, ]

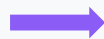
cat(sum(keep), "of", nrow(counts),
    "genes retained\n") # log this number!
```

Why filter at all? Near-zero genes carry no biological signal but inflate the multiple-testing burden and slow every downstream step.

Thresholds are judgment calls. Match the minimum sample count to your smallest group size; justify the choice in one sentence.

Typical Effect

Before: 58,000 genes



After: 17,500 genes

70% removed — the signal is unchanged, the noise is gone.

Sample-level QC screens every sample before you trust the matrix

1 Library Size

Total reads per sample (colSums).

A sample with 10x fewer reads than its peers is a technical failure candidate.

2 Distribution Shape

Density plot of log counts per sample.

One curve shifted far from the bundle signals a processing artifact.

3 Boxplot Spread

Per-sample boxplots of log expression.

Misaligned medians reveal samples needing normalization — or removal.

The QC Mindset

QC does not delete data — it flags evidence. Every exclusion needs a **documented, defensible reason**.

PCA is the fastest way to see whether biology or batch drives your data

Bad: samples cluster by batch



● Batch 1 (mixed groups)

◆ Batch 2 (mixed groups)

PC1 separates sequencing date, not condition

Good: samples cluster by group



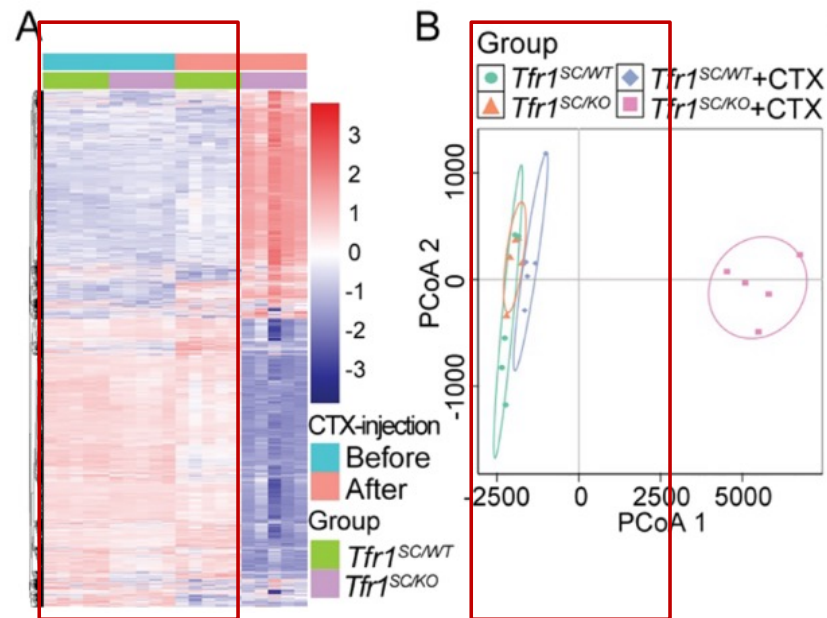
● Control (both batches)

◆ Treatment (both batches)

PC1 separates condition; batches interleave

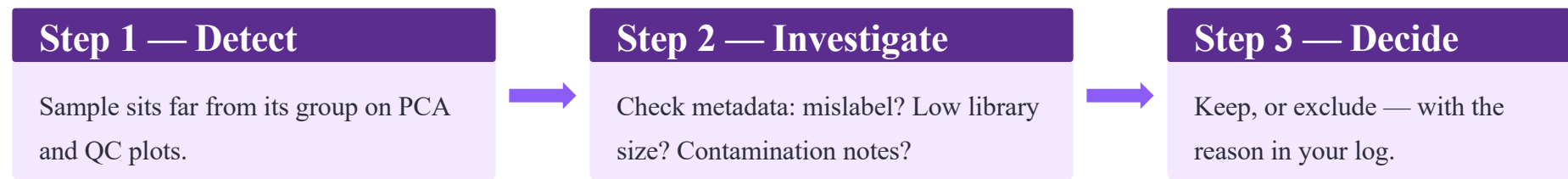
Always color the **same PCA plot** by group, then by batch — the comparison is the diagnosis.

PCA is the fastest way to see whether biology or batch drives your data



An outlier sample is evidence to investigate, not automatically trash

The Outlier Decision Ladder



Keep when: the outlier is biological (true responder, real heterogeneity) and group balance survives its inclusion.

Exclude when: there is a technical explanation — failed library, mislabeled tube, contaminated lane — recorded in writing.

Removing samples to make a result **prettier** is data fabrication. Removing them with **documented technical cause** is good science.

Handle batch in the design model first; reach for ComBat only when needed

Option A — Include batch in the model

Add batch as a covariate in the design formula.

The raw counts stay untouched; the model estimates and removes batch statistically.

Option B — Correct with ComBat

Adjust the matrix itself (sva::ComBat).

Use for visualization and clustering — never as input to count-based DE tools.

```
# Option A: model-based (preferred)
design <- ~ batch + condition
# Option B: for plots only
adj <- sva::ComBat(log1p(counts), batch = batch)
```

Key rule: DESeq2 wants raw counts plus a design formula — feed it Option A, keep ComBat for figures.

Normalization makes samples comparable — each method answers a different question

Method	What It Corrects	Appropriate Use	Watch Out
Raw counts	Nothing — sequencing output as-is	Input for DESeq2 / edgeR	Not comparable across samples
CPM / TPM	Library size (gene length for TPM)	Quick plots, within-study views	Not for DE testing
DESeq2 size factors	Sequencing depth and composition	Differential expression	Assumes most genes unchanged
VST / rlog	Depth plus variance stabilization	PCA, clustering, heatmaps	Transformed — not counts

One pipeline, two matrices: raw counts for statistics, transformed values for visualization — never swap their roles.

DESeq2 requires raw counts — TPM from GEO is a dead end for DE analysis

Why the Model Needs Integers

DESeq2 and edgeR model counts with a **negative binomial** distribution.

The model learns the mean–variance relationship from integer data — TPM and FPKM destroy that structure.

The Classic Trap

Downloading the GEO Series Matrix often gives you the **author-normalized** values.

Rounding TPMs to integers does not recover counts — it only hides the problem.

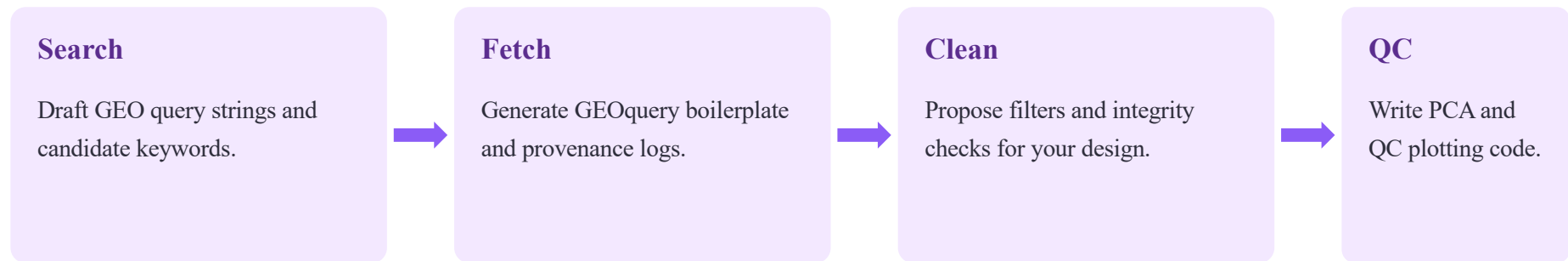
Where to Find Real Counts

1. Supplementary files on the GSE page (counts.txt, .csv, .tsv).

2. Recount3 / ARCHS4 — uniformly reprocessed count matrices.

3. Re-quantify from SRA fastq files (advanced, most reproducible).

AI accelerates every pipeline stage — if you keep it on a short leash



Three Non-Negotiable Guardrails

Verify every claim. AI invents GEO accessions, functions, and column names — check each one in R.

Never paste data blindly. Run AI code line by line on a small subset before the full matrix.

Log the conversation. Prompts and accepted suggestions belong in logs/ — part of provenance.

Ask AI for a plan before you ask it for code

Weak Prompt — "code first"

"Write R code to download and clean GSE158422."

Result: plausible code, wrong assumptions, no checks.

Strong Prompt — "plan first"

"Before any code: list the cleaning steps, the checks after each step, and what could go wrong — for GSE158422."

Why Plan-First Wins

You stay the scientist. You approve the strategy; the model only accelerates execution.

Errors surface early. A wrong plan is cheap to fix; wrong code run on real data is not.

The plan becomes documentation. Paste the approved plan straight into your log.

Activity: find the three silent bugs hiding in this AI-generated code

ai_draft.R – looks fine, is not

```
gse <- getGEO("GSE158422")[[1]]
mat <- exprs(gse)                # bug 1?
meta <- pData(gse)

# merge a clinical table by sample name
full <- merge(meta, clinical,
              by = "title")      # bug 2?

mat_r <- round(mat)              # bug 3?
dds <- DESeqDataSetFromMatrix(mat_r, ...)
```

Work in Pairs — 8 Minutes

Hint 1. What data level does the GEO Series Matrix usually contain — and what does DESeq2 need?

Hint 2. What does merge() silently do to row order and unmatched samples?

Hint 3. Does round() turn normalized values back into real counts?

This week: fetch, clean, and document one real GEO dataset

Homework — The Full Mini-Pipeline

1. Choose a GEO dataset with raw counts available.
2. Complete the dataset triage scorecard.
3. Fetch it with GEOquery; build the project folder.
4. Run all five integrity checks; filter genes.
5. Produce QC plots plus one PCA colored by group and by batch.
6. Write a one-page provenance report: every decision, every threshold, every AI prompt used.

Quick Quiz — 3 Questions

1. GEO Series Matrix gives TPM — can you run DESeq2?
2. PCA clusters by date, not group. Diagnosis?
3. Name two integrity checks that run in seconds.

Next Week — Week 4

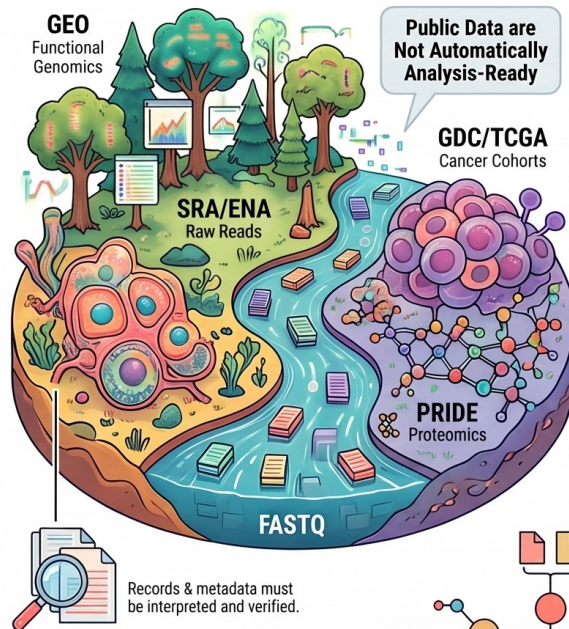
From clean counts to biological answers:

Differential expression with DESeq2, multiple testing, and your first volcano plot.

THE BIOINFORMATICS DATA PIPELINE: FROM RESEARCH QUESTION TO ANALYSIS-READY DATA

THE PUBLIC REPOSITORY ECOSYSTEM

Select the Repository by Assay, Not Ease of Download



Open vs. Controlled Access

Publicly described

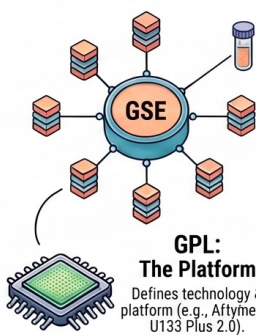
Human genomic data often requires institutional approval



GEO ACCESSION GRAMMAR & LITERACY

GSE: The Series

Primary entry point, often spans multiple platforms.



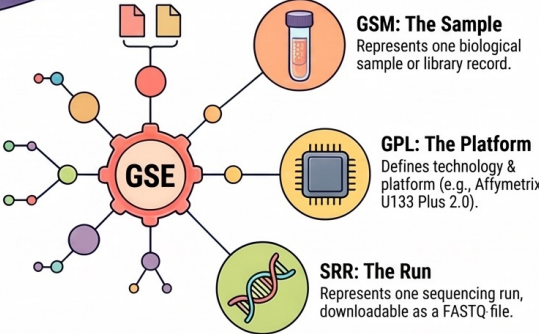
GPL: The Platform

Defines technology & platform (e.g., Affymetrix U133 Plus 2.0).

GEO ACCESSION GRAMMAR & LITERACY

GSE: The Series

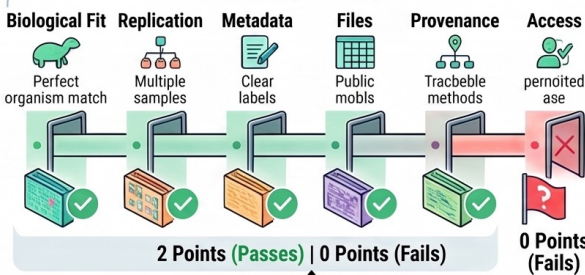
Primary entry point, often spans multiple platforms.



THE 12-POINT DATASET TRIAGE

Score Before You Store

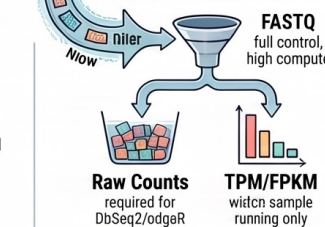
0-2 scale across six criteria



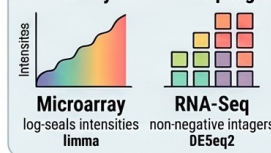
The "9/12" Decision Rule:
Proceed only when total score is ≥ 9 out of 12

DATA LAYERS & FILE FORMATS

FASTQ vs. Raw Counts vs. TPM/FPKM



Microarray vs. RNA-Seq Logic



THE "SILENT FAILURE" CHECK

Explicitly Align Metadata to Matrix

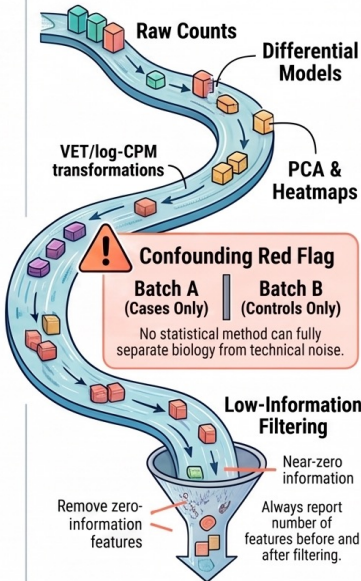
	C1	C2	C3	Metadata
R1				
R2				
R3				

R commands (e.g., `stopifnot(identical(colnames(expr), rownames(metadata)))`) ensure expression columns and metadata rows refer to the same samples.

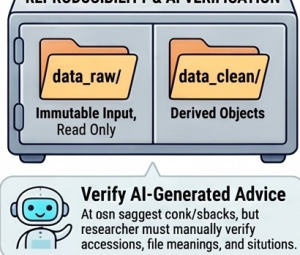
Beware of Misaligned Rows
6 may still run, producing valid looking—but biologically meaningless—results.

CLEANING, QC, AND BATCH EFFECTS

Normalization is Purpose-Specific



REPRODUCIBILITY & AI VERIFICATION



We will have quiz next week.

Do not forget to review the class material.

5 multiple choice questions

1 min of each question for total 5 mins

Data Fetching and Cleaning from Public Repositories

Week 3 fetch GEO, NCBI and Reproducible R Workflows

Lab Samples

Sequencing

Public Repository

Your Analysis

New Discovery

Pre-Reading for Week 4

From DNA measurement to FASTQ, alignment, variant calling, epigenetic modification,
transcription factor binding and evidence aware interpretation

Week 3 Complete